Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

At infinite dilution, the percentage ionization for both strong and weak electrolyte is

(A) 1%

(B) 20%

(C) 50%

(D) 100%

Q.2

Which of the following can act both as Bronsted acid and Bronsted base

(A) CI

(B) HCO_3

(C) H_3O^+

(D) OH

Q.3

Ag_2CrO_4 and Ag_2CO_3 are simultaneously dissolved in water. The solubility of $Ag_2CrO_4 \& Ag_2CO_3$ are x' and $'y'$ mo M respectively. If the K_{ep} of Ag_2CrO_4 is 'a' and that of Ag_2CO_3 is 'b', then which of the following is correct?

(A) $(x+y)^2x = n$

(B) $(2x+2y)y = b$

(C) $(2x+2y)^2x = b$

(D) $(2y+2x)^2y = b$

Q.4

Which equilibrium can be described as an acidbase reaction using the Lewis acid-base definition but not using the Bronsted-Lowry definition?

(A) $2NH_3 + H_2SO_4 \rightleftharpoons 2NH_4^+ + SO_4^{2-}$

(B) $NH_3 + CH_3COOH \rightleftharpoons NH_4^+ + CH_3COO^-$

(C) $H_2O + CH_3COOH \rightleftharpoons H_3O^+ + CH_3COO^-$

@IITJEE_Advanced

(D) $[Cu(H_2O)_4]^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+} + 4H_2O$

Q.5

At a certain temperature the dissociation constants of formic acid and acetic acid are 1.8×10^{-4} and 1.8×10^{-6} respectively. The concentration of acetic acid solution in which the hydrogen ion has the same concentration as in $0.001M$ formic acid solution is equal to

(A) $0.001M$

(B) $0.01M$

(C) $0.1M$

(D) $0.0001M$

Q.6

What is the decreasing order of basic strengths of $OH^-, NH_2^-, H-C\equiv C^-$ and $CH_3-CH_2^-$

(A) $CH_3-CH_2^- > NH_2^- > H-C\equiv C^- > OH^-$

(B) $H-C\equiv C^- > CH_3CH_2^- > NH_2^- > OH^-$

(C) $OH^- > NH_2^- > H-C\equiv C^- > CH_3-CH_2^-$

(D) $NH_2^- > H-C\equiv C^- > OH^- > CH_3-CH_2^-$

Q.7

When CO_2 dissolves in water, the following equilibrium is established $CO_2 + 2H_2O \rightleftharpoons H_3O^+ + HCO_3^-$
for which the equilibrium constant is 3.8×10^{-7} and $pH = 6.0$. The ratio of $[HCO_3^-]$ to $[CO_2]$ would be

(A) 3.8×10^{-13}

(B) 3.8×10^{-1}

(C) 6.0

(D) 13.4

Q.8

Values of dissociation constant, K_a are given as follows:

Acid	K_a
HCN	6.2×10^{-10}
HF	7.2×10^{-4}
HNO_2	4.0×10^{-4}

Correct order of increasing base strength of the base CN^-, F^- and NO_2^- will be :

@IITJEE_Advanced

(A) $F^- < CN^- < NO_2^-$

(B) $NO_2^- < CN^- < F^-$

(C) $F^- < NO_2^- < CN^-$

(D) $NO_2^- < F^- < CN^-$

Q.9

Given

(i) $HCN(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CN^-(aq)$

$K_a = 6.2 \times 10^{-10}$

(ii) $CN^-(aq) + H_2O(l) \rightleftharpoons HCN(aq) + OH^-(aq)$

$K_b = 1.6 \times 10^{-5}$

These equilibria show the following order of the relative base strength,

(A) $OH^- > H_2O > CN^-$

(B) $OH^- > CN^- > H_2O$

(C) $H_2O > CN^- > OH^-$

(D) $CN^- > H_2O > OH^-$

Q.10

The degree of dissociation of $0.1M$ weak acid HA is 0.5%. If $2\ mL$ of $1.0MHA$ solution is diluted to $32\ mL$ the degree of dissociation of acid and H_3O^+ ion concentration in the resulting solution will be respectively

(A) 0.02 and 3.125×10^{-4}

(B) 1.25×10^{-3} and 0.02

(C) 0.02 and 1.25×10^{-3}

(D) 0.02 and 8.0×10^{-12}

Q.11

The conjugate base of hydrazoic acid is:

(A) N^{-3}

(B) N_3^-

(C) N_2^-

(D) HN_3^-

Q.12

A solution of $0.1MNaZ$ has $pH = 8.90$. The K_a of HZ is

@IITJEE_Advanced

(A) 6.3×10^{-11}

(B) 6.3×10^{-10}

(C) 1.6×10^{-5}

(D) 1.6×10^{-6}

Q.13

If SO_2 content in the atmosphere is $0.12 ppm$ by volume, pH of rain water is (assume 100% ionisation of acid rain as monobasie acid)

(A) 5.7

(B) 5.6

(C) 5.4

(D) 2.0

Q.14

The hydrogen ion concentration of 0.2 N CH_3COOH which is 40% dissociated is

(A) 0.08 N

(B) 0.12 N

(C) 0.80 N

(D) 1.2 N

Q.15

The first and second dissociation constants of an acid H_2 A are 1.0×10^{-5} and 5.0×10^{-10} respectively. The overall dissociation constant of the acid will be

(A) 0.2×10^{5}

(B) 5.0×10^{-5}

(C) 5.0×10^{15}

(D) 5.0×10^{-15}

Answers & Solutions

Q.1 Answer:

100%

Solution:

According to ostwald's dilution law because degree of ionization is directly proportional to ionization.

@IITJEE_Advanced

Q.2 Answer:

HCO_3

Solution:

Those substance which accepts a proton are called Bronsted base and which donate a proton are called Bronsted acid

$HCO_3^- + H^+ \rightleftharpoons H_2CO_3$ **Bronsted base**

$HCO_3^- \rightleftharpoons H^+ + CO_3^-$ **Bronsted acid**

Q.3 Answer:

$(2y + 2x)^2 y = b$

Solution:

$Ag_2CrO_4(s) \rightleftharpoons 2Ag^+ + CrO_4^{2-}$

$(2x + 2y) \quad x$

$Ag_2CO_3(s) \rightleftharpoons 2Ag^+ + CO_3^{2-}$

$(2x + 2y)y$

K_{ap} **of** $Ag_2CrO_4 = a = (2x + 2y)^2 x$

K_{gp} **of** $Ag_2CO_3 = b = (2x + 2y)^2 y$

Q.4 Answer:

$[Cu(H_2O)_4]^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+} + 4H_2O$

Solution:

$[Cu(H_2O)_4]^{2-} + 4NH_3 \leftrightharpoons [Cu(NH_3)_4]^{2+} + 4H_2O$

According to Bronsted-Lowry concept of acids and bases, an acid is a substance which can give a proton and a base is a substance which accepts a proton.

According to Lewis concept of acids and bases, an acid is a substance which can accept a lone pair of electrons whereas a base is a substance which can donate a lone pair of electrons.

$[Cu(H_2O)_4]^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+} + 4H_2O$ **involves loss and gain of electrons.**

H_2O **is coordinated to** Cu **by donating electrons (LHS). It is then removed by withdrawing electrons.**

Q.5 Answer:

$0.1M$

Solution:

$[H^+] = \sqrt{C \times K_a} = \sqrt{0.001 \times 1.8 \times 10^{-4}}$ **for\r**

formic acid\r

$[H^+] = \sqrt{C_2 \times 1.8 \times 10^{-6}}$ **for acetic acid\r**

Equating and solving, $C_2 \times 1.8 \times 10^{-6} = 10^{-3} \times 1.8 \times 10^{-4}$

$C_2 = 1\ M$

@IITJEE_Advanced

Q.6 Answer:

$CH_3 - CH_2^- > NH_2^- > H - C \equiv C^- > OH^-$

Solution:

Follow Lowry Bronsted concept for conjugate pair. Acid strength follows the order $H_2O > C_2H_2 > NH_3 > C_2H_6$ **conjugate base will follow the order**

$C_2H_5^- > NH_2^- > C_2H^- > OH^-$

Q.7 Answer:

3.8×10^{-1}

Solution:

$M_1\ V_1 = M_2\ V_2$

or $2 \times 1 = 32 \times M_2$

Hence, $M_2 = \frac{1}{16}$

For this diluted solution, $\alpha = \sqrt{\frac{K_a}{C}} = 0.632\%$.

The hydrogen ion concentration is $[H^+] = c\alpha = \frac{1}{16} \times 0.00632 = 3.955 \times 10^{-4}$.

Q.8 Answer:

$F^- < NO_2^- < CN^-$

Solution:

Higher the value of K_a **lower will be the value of** pK_a **i.e. higher will be the acidic nature. Further** CN^-, F^- **and** NO_2^- **are conjugate base of the acids** HCN, HF **and** HNO_2 **respectively hence the correct order of base strength will be**

$$F^- < NO_2^- < CN^-$$

$(\because$ **stronger the acid weaker will be its conjugate base)**

Q.9 Answer:

$OH^- > CN^- > H_2O$

Solution:

The more is the value of equilibrium constant, the more is the completion of reaction or more is the concentration of products i.e. the order of relative strength would be

$$OH^- > CN^- > H_2O$$

Q.10 Answer:

0.02 and 1.25×10^{-3}

@IITJEE_Advanced

Solution:

$\alpha_1 = 0.005 = \sqrt{K_a \times C_1}$

Molarity of diluted solution; $2 \times 1 = 32 \times M$

$M = \frac{1}{16}(C_2)$

$\alpha_2 = \sqrt{\frac{K_a}{C_2}} = 0.005\sqrt{16} = 0.02$

$H_3O^+ = C_2\alpha_2 = \frac{0.02}{16} = 1.25 \times 10^{-3}M$

Q.11 Answer:

N_3^-

Solution:

$N_3H \quad \rightleftharpoons N_3^- + H^+$

Hydrazoic acid

i.e, conjugate base of hydrazoic acid is N_3^-

Q.12 Answer:

1.6×10^{-5}

Solution:

NaZ **is salt of** W_A/S_B

$\therefore pH = \frac{1}{2}(pK_w + pK_a + \log C)$

$8.9 \times 2 = 14 + pK_a + \log 0.1$

$17.8 = 14 + pK_a - 1$

$pK_a = 4.8$

$K_a = \text{Antilog}(-4.8)$

$= 1.585 \times 10^{-5} \approx 1.6 \times 10^{-5}$

Q.13 Answer:

5.7

Solution:

$0.12\text{ppm} = 0.12 \text{ g in } 10^6 \text{ mL}$

$[H^+] = [SO_2]$

$\therefore H_2SO_3] = \frac{0.12}{64} \times 10^{-3}M$

$\mathbf{pH} = 5.7$

Q.14 Answer:

0.08 N

Solution:

@IITJEE_Advanced

$[\mathrm{H}^+] = \mathrm{C}\alpha = 0.2 \times 0.40 = 0.08\ \mathrm{N}$

$$\left(\alpha = \frac{40}{100} = 0.40\right)$$

Q.15 Answer:

5.0×10^{-15}

Solution:

$\mathrm{H_2\ A} \rightleftharpoons \mathrm{H^+} + \mathrm{HA^-}$

$\therefore K_1 = 1.0 \times 10^{-5} = \frac{[\mathrm{H^+}][\mathrm{HA^-}]}{[\mathrm{H_2\ A}]}$ **(Given)**

$\mathrm{HA^-} \longrightarrow \mathrm{H^+} + \mathrm{A^{2-}}$

$\therefore K_2 = 5.0 \times 10^{-10} = \frac{[\mathrm{H^+}][\mathrm{A^{2-}}]}{[\mathrm{HA^-}]}$ **(Given)**

$$= \frac{[\mathrm{H^+}]^2\,[\,\mathrm{A^{2-}}]}{[\mathrm{H_2\ A}]} = K_1 \times K_2$$
$$= \left(1.0 \times 10^{-5}\right) \times \left(5 \times 10^{-10}\right) = 5 \times 10^{-15}$$

@IITJEE_Advanced